

POPURI VAMSI KRISHNA

Milpitas, CA | 408-592-4482 | vamsipopuri1997@gmail.com | [linkedin.com/in/vamsipopuri](https://www.linkedin.com/in/vamsipopuri) | github.com/Vamsipops

SKILLS

Programming & Development	Python, Java, Flask (API Development), Fast API, Streamlit, React, Node.js
Databases & Search	MySQL, Elasticsearch, ChromDB, Pinecone, MongoDB
Generative AI & Data Science	RAG, vector search, prompt engineering, document chunking, embeddings, Tableau
Cloud & Enterprise Tools	AWS (Lambda, LightSail, RDS, S3, CloudWatch, SNS, SES), Docker, Google APIs, ServiceNow, Jira
Data Engineering	Fact and Dimension table design, Data Transformation & Migration, Micro-services Architecture
Interpersonal Skills	Team Player, Empathetic, Grit, Conflict Resolution

WORK EXPERIENCE

Associate Director/Software Engineer | Lexim AI, Cupertino, California, USA Sep 2023 – Jan 2025

- Architected and implemented **Retrieval-Augmented Generation (RAG)** for document processing, integrating **OpenAI** and **Elasticsearch** to enhance data retrieval efficiency.
- Designed up to 200 **REST APIs** using **Flask** and **Fast API** to power application functionalities within a microservices architecture.
- Containerized** and deployed 12 Python applications on **AWS LightSail** using **Docker**, ensuring scalable and resilient cloud-native deployments.
- Optimized **MySQL** data models, designing robust Fact and Dimension tables to streamline query performance in data warehousing scenarios improving dashboard loading time by 50%
- Enhanced **full-text search** using **Elasticsearch**, reducing search times from 3–4 seconds to under 1 second
- Automated key data workflows using **AWS Lambda**, reducing manual intervention and operational overhead by 90%
- Managed **CI/CD pipelines** and monitored applications with **AWS CloudWatch**, **SNS**, and **SES**.
- Utilized **React** to implement critical front-end functionalities (login, logout, and home page) ensuring a cohesive full stack solution.

Associate Systems Engineer | TCS, Kolkata, West Bengal, India May 2019 – Jan 2025

- Managed SAP Billing application support and development for a Canada-based utility customer (ENMAX) using Agile methodology and Test-Driven Development (TDD), improving enhancement delivery efficiency
- Developed more than 50 SAP ABAP enhancements in collaboration with cross-functional teams, tailored to dynamic business requirements using the iterative Agile development approach, resulting in improved system functionality and user satisfaction
- Utilized ServiceNow for effective incident management across six cross-functional teams, creating dashboards to foster efficient incident resolution and reporting.

EDUCATION

Master of Science in Data Analytics Aug 2021 – May 2023

San Jose State University, San Jose, California, USA.

Bachelor of Technology in Mechanical Engineering Aug 2015 – April 2019

Gokaraju Rangaraju Institute of Technology, Hyderabad, India.

PROJECTS

Design and Manufacturing of All-Terrain Vehicle (SAE BAJA) Aug 2016 - Feb 2019

- Led the design, manufacturing, and assembly of an All-Terrain Vehicle meeting competition requirements.
- Progressed from junior member to club captain over three years, taking on leadership and strategic planning roles
- Managed a project budget of \$20,000–\$25,000, overseeing material procurement and engaging with over 20 vendors for machining and custom parts manufacturing.
- Participated in six national SAE competitions, achieving a notable second-place finish in one event

ACHIEVEMENTS & VOLUNTEERING

- Achieved first place at the department level in the Smart India Hackathon with a project focused on a Negative Pressure Irrigation System.
- Volunteered for election duty during the Andhra Pradesh General Election in 2015.